



# **AUTOMATED SKIN DISEASE DIAGNOSIS AND TREATMENT RECOMMENDATIONS**



## **A DESIGN PROJECT REPORT**

*Submitted by*

**AJAY KUMAR S**

**DINESH M**

**VIGNESH S**

*in partial fulfillment for the award of the degree*

*of*

**BACHELOR OF TECHNOLOGY**

*in*

**ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**

**K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**

(An Autonomous Institution, affiliated to Anna University Chennai and Approved by AICTE, New Delhi)

**SAMAYAPURAM – 621 112**

**JUNE, 2025**



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**K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**  
**(AUTONOMOUS)**  
**SAMAYAPURAM – 621 112**

**BONAFIDE CERTIFICATE**

Certified that this design project report titled “**AUTOMATED SKIN DISEASE DIAGNOSIS AND TREATMENT RECOMMENDATIONS**” is the bonafide work of **AJAY KUMAR S (811722243002), DINESH M (811722243014), VIGNESH S (811722243053)**, who carried out the design project under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form part of any other design project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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**INTERNAL EXAMINER**

**EXTERNAL EXAMINER**

## DECLARATION

We jointly declare that the design project report on “**AUTOMATED SKIN DISEASE DIAGNOSIS AND TREATMENT RECOMMENDATIONS**” is the result of original work done by us and best of our knowledge, similar work has not been submitted to “**ANNA UNIVERSITY CHENNAI**” for the requirement of Degree of **BACHELOR OF TECHNOLOGY**. This design project report is submitted on the partial fulfilment of the requirement of the award of Degree of **BACHELOR OF TECHNOLOGY**.

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## **ABSTRACT**

Skin diseases are one of the most common health issues affecting people across different age groups. Early identification and proper treatment are crucial to avoid complications. This project presents an automated solution that combines computer vision and generative AI to support skin disease diagnosis and explanation. A deep learning model (MobileNetV2) is used to classify the type of skin disease from an input image. After classification, a Generative AI model (LLaMA 3 from Ollama) provides a detailed yet easy-to-understand explanation of the disease, its causes, and treatment suggestions. The system is designed to be fast, lightweight, and user-friendly, making it suitable for mobile or web platforms. This project aims to assist both common users and healthcare professionals with early skin disease detection and awareness..

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## **LIST OF ABBREVIATIONS**

|          |                                              |
|----------|----------------------------------------------|
| CNN      | - Convolutional Neural Network               |
| GPU      | - Graphical Processing Unit                  |
| Grad-CAM | - Gradient-weighted Class Activation Mapping |
| GUI      | - Graphical User Interface                   |
| LLM      | - Large language model                       |
| ReLU     | - Rectified Linear Unit                      |
| RGB      | - Red, Green, Blue                           |
| ROI      | - Region of Interest                         |
| SGD      | - Stochastic Gradient Descent                |

## **CHAPTER 1**

### **INTRODUCTION**

#### **1.1 OVERVIEW**

Skin diseases are among the most widespread health concerns, affecting people of all ages around the world. Early detection and proper treatment are vital to prevent complications, but limited access to dermatologists—especially in remote or underserved areas—can delay care. This project presents an AI-driven solution that combines computer vision and generative AI to help detect and explain common skin conditions. The system uses MobileNetV2, a lightweight convolutional neural network (CNN), to classify skin diseases based on input images. Trained on a labeled dataset, the model can identify a range of conditions such as eczema, psoriasis, and melanoma with high accuracy. Its efficiency makes it ideal for mobile or web deployment, allowing for real-time classification even on low-power devices.

Once a skin condition is identified, the prediction is passed to LLaMA 3, a local large language model integrated through Ollama. This model generates a clear, user-friendly explanation of the disease, covering its causes, symptoms, and general treatment options. The integration of generative AI ensures that users not only receive a diagnosis but also understand what it means in practical terms. Importantly, the entire system can run locally, preserving user privacy and enabling offline use. This feature makes it especially useful in areas with limited internet access or in privacy-sensitive environments. The system is designed to be intuitive, accessible, and informative—supporting both individuals seeking self-assessment and professionals in need of quick AI-assisted insights. It can also serve as an educational tool for increasing awareness of skin health. In summary, this project demonstrates how a combination of deep learning and generative AI can provide accurate, interpretable, and privacy-conscious assistance for skin disease detection and explanation, offering a promising solution for AI-assisted healthcare applications.

## 1.2 OBJECTIVE

The primary objective of this project is to develop an AI-powered system that aids in the early detection and understanding of common skin diseases by integrating computer vision and generative AI technologies. The system functions as a supportive tool for both individuals and healthcare providers, offering quick, informative, and privacy-conscious assistance. At its core, it uses MobileNetV2, a lightweight yet powerful convolutional neural network (CNN), to analyze and classify skin conditions from user-submitted images. This model has been fine-tuned on a curated dermatological dataset to accurately recognize various conditions such as eczema, psoriasis, and melanoma. Its efficiency and speed make it suitable for real-time use on mobile or web platforms.

To complement the classification process, the system incorporates LLaMA 3, a local large language model accessed via Ollama, to generate simple and understandable explanations of the predicted disease. These explanations include essential information such as common symptoms, potential causes, and general treatment recommendations. This combination allows users to receive not only a prediction but also educational insight into the condition, helping bridge the gap between diagnosis and awareness. The project places strong emphasis on accessibility, ethical use, and informed decision-making. By enabling offline usage and maintaining user privacy, it is particularly suited for deployment in rural or underserved areas. Additionally, clear disclaimers are included to highlight that the system is not a replacement for medical advice, but rather a tool to encourage early awareness and prompt consultation with healthcare professionals. Overall, the objective is to provide a user-friendly, efficient, and responsible AI solution that supports skin health literacy and promotes timely medical intervention

## **CHAPTER 2**

### **LITERATURE SURVEY**

#### **2.1 AUTOMATIC SKIN DISEASE DIAGNOSIS USING DEEP LEARNING FROM CLINICAL IMAGE AND PATIENT INFORMATION**

**Oluwashina Oyeniran, Joshua Oyeniyi**

This paper presents a multimodal approach to improve skin disease diagnosis by combining image-based analysis with patient information. The system is built using integrated learning models that mimic real clinical workflows. It supports more realistic medical outcomes by processing both types of data in parallel. However, the methodology is not clearly detailed and performance results are not thoroughly verified.

##### **Merits:**

- Combines clinical images and patient metadata, offering a more holistic diagnostic approach.
- Closely mimics actual clinical workflows, making the model more relevant to real-world settings.
- Introduces an innovative multimodal integration that enhances diagnostic accuracy.

##### **Demerits:**

- Lacks clear implementation details, making replication or improvement difficult.
- The paper is vague in structure, lacking technical depth in methodology explanation.
- No robust accuracy metrics or performance validations are provided.

## **2.2 AN AUTOMATED SKIN DISEASE DIAGNOSIS SYSTEM**

**Lawrence Omotosho, Kehinde Sotinwa, Benjamin Adegoke**

This study proposes a CNN-based approach using AlexNet and transfer learning to classify skin diseases. The model is trained on real medical images sourced from clinical environments in Nigeria. It provides accurate classification for common skin conditions. However, the model is trained on region-specific data, which limits its generalization to other populations.

### **Merits:**

- Achieves high classification accuracy using AlexNet with transfer learning, ensuring reliable skin disease predictions.
- Utilizes real-world clinical images, increasing the model's relevance and applicability in actual medical settings.
- Transfer learning reduces training time and resource needs, making the approach suitable for low-compute environments.

### **Demerits:**

- Dataset is region-specific, restricting the model's effectiveness outside the local demographic context.
- Performance may degrade on diverse populations due to limited exposure to varying skin tones and conditions.
- Scalability is limited, as retraining with more diverse data is required for deployment in broader regions.

## **2.3 APPLICATIONS OF AI IN DERMATOLOGY**

**Anna Escalé-Besa, Josep Vidal-Alaball, Aina Fuster-Casanovas**

This paper explores how different AI models like ANN, CNN, and DNN are being used in dermatology. It highlights the ability of these models to reduce unnecessary clinical referrals and improve diagnostic accuracy. The focus is on AI's role in supporting early screening and treatment recommendation. However, the paper mentions that many AI models are trained on theoretical image data without sufficient real-world validation.

### **Merits:**

- Demonstrates how AI can accelerate diagnosis, helping reduce unnecessary clinical appointments and workload.
- Highlights AI's value in early screening, enabling quicker medical intervention and better patient outcomes.
- Shows potential of AI to support dermatologists in routine decision-making and reduce diagnostic errors.

### **Demerits:**

- Many AI models lack real-world validation, raising concerns about their effectiveness in clinical deployment.
- Performance may decline when tested on complex or atypical cases not represented in training data.
- Heavy reliance on theoretical datasets may lead to overfitting and reduced reliability in real settings.

## 2.4 GENERATIVE AI IN IMPROVING PERSONALIZED CARE PLANS

**Mirza Mansoor Baig, Chris Hobson, Hamid GholamHosseini, Ehsan Ullah**

This study discusses how generative AI, especially large language models, can create personalized treatment plans for patients based on medical profiles. The model customizes care plans and supports patient-centered healthcare. However, the transparency of these models is a challenge, and their outputs can sometimes be hard to interpret.

### **Merits:**

- Personalizes treatment plans based on individual medical profiles, improving the relevance of care.
- Enhances patient-centered healthcare by tailoring advice to specific symptoms and conditions.
- Utilizes generative AI to automate and adapt care plans dynamically as patient data evolves.

### **Demerits:**

- Model decisions can be hard to interpret, making clinical validation and trust a challenge.
- There's a risk of inconsistent or inaccurate outputs without proper medical oversight.
- Lack of transparency may reduce user confidence in AI-generated treatment suggestions.

## **2.5 GEN AI-POWERED CLINICAL GUIDELINES: AUTOMATING THE FUTURE OF EVIDENCE-BASED CARE**

**Ramankar Reddy Dunda**

This paper introduces a generative AI system that automates the creation of clinical guidelines by using electronic health records (EHR) and predictive analytics. It also incorporates technologies like edge computing and 5G for immersive AI integration. The model reduces clinician workload by providing AI-generated decisions and insights. However, the system relies heavily on high-quality data and may lack transparency in how the guidelines are produced.

### **Merits:**

- Automates the generation of clinical guidelines, significantly reducing manual effort for healthcare providers.
- Integrates predictive analytics and EHR data to offer timely, evidence-based medical recommendations.
- Enhances decision-making efficiency by streamlining access to personalized clinical insights.

### **Demerits:**

- Lack of transparency in the model's reasoning can reduce trust in AI-generated medical advice.
- System performance heavily depends on the quality, accuracy, and completeness of input medical data.
- Errors in data or prediction may lead to flawed or unsafe clinical guidelines without proper validation.

## **CHAPTER 3**

### **SYSTEM ANALYSIS**

#### **3.1 EXISTING SYSTEM**

In traditional medical practice, the diagnosis of skin diseases is typically performed through direct clinical examination conducted by qualified dermatologists. This process heavily relies on the specialist's ability to visually assess the affected area based on features such as color, texture, shape, and the spread of skin abnormalities. Doctors also depend on their medical training, experience, and judgment to identify specific dermatological conditions. When visual inspection alone is not sufficient, supplementary diagnostic tools such as dermoscopy devices are used to magnify skin lesions for closer examination. In more complex or potentially cancerous cases, a biopsy may be taken to analyze skin tissue at a microscopic level. These traditional diagnostic techniques are well-established and medically validated, offering accurate results when carried out by experienced professionals.

However, this conventional method poses several limitations, especially in today's healthcare environment where speed, consistency, and accessibility are critical. Manual diagnosis often requires patients to physically visit hospitals or clinics, sometimes enduring long wait times for appointments or test results. In many public healthcare systems, overcrowding can extend the time needed to receive a definitive diagnosis, which may negatively impact patient care. Moreover, the accuracy of this approach is largely dependent on the examining doctor's experience and interpretation, which may introduce inconsistencies between different practitioners. This subjectivity can lead to either overdiagnosis or missed diagnoses, especially for rare or atypical cases. Access to dermatologists is also limited in rural or underserved areas, where patients might face geographic or economic barriers. These challenges result in delayed treatments, reduced diagnostic quality, and unequal access to care. Therefore, there is a growing need for automated, fast, and reliable systems that can aid in skin disease diagnosis, particularly in resource-limited settings where expert consultation may not always be feasible.

### 3.1.1 Demerits:

- **Time-Consuming Process**

Diagnosis often involves multiple stages such as consultation, physical examination, and lab testing, which can delay treatment.

- **Limited Accessibility**

Dermatologists are often concentrated in urban hospitals, leaving rural or underdeveloped areas underserved.

- **Subjective Judgments**

Diagnostic accuracy may vary based on the experience, fatigue, and perception of individual healthcare providers.

- **Overburdened Healthcare Infrastructure**

Manual diagnosis contributes to long patient queues and overextended doctors in public healthcare systems.

- **High Cost of Diagnostic Procedures**

Advanced tests like biopsies and dermoscopic imaging may be expensive and unaffordable for economically weaker populations.

- **Delayed Detection of Rare Conditions**

Less common skin diseases may go unrecognized in early stages due to limited exposure or knowledge among general practitioners.

- **Lack of Digitized Records and Automation**

Traditional methods often lack integration with digital tools, reducing efficiency in follow-up, documentation, and knowledge sharing.

### **3.2 PROPOSED SYSTEM**

The proposed system introduces an AI-based diagnostic platform aimed at enhancing skin disease detection using image classification and natural language generation. At its core, the system utilizes MobileNetV2, a lightweight and efficient Convolutional Neural Network (CNN) trained on large datasets of skin disease images. This model can accurately classify skin conditions by analyzing features such as color, shape, and texture from user-submitted photographs. Once the classification is completed, the disease label is forwarded to LLAMA 3, a state-of-the-art large language model accessed locally via Ollama. LLAMA 3 generates comprehensive explanations that include the disease's characteristics, potential causes, symptoms, and general treatment recommendations in simple, user-friendly language.

The entire diagnostic process is wrapped in a user-accessible web interface built with Gradio, which simplifies the user experience. Users can upload an image of the affected skin area, and within seconds, receive a possible diagnosis along with informative text. This approach removes barriers associated with conventional systems, such as long wait times, travel requirements, and the availability of specialists. In regions with limited access to dermatologists or medical infrastructure, this system provides a much-needed first step in healthcare engagement. Moreover, since the analysis and report generation are fully automated and processed locally, it ensures user privacy and minimizes data sharing risks. While not a replacement for clinical judgment, this AI-powered assistant enhances early awareness, helps prioritize cases that need professional attention, and promotes proactive skin health management. By combining deep learning and generative AI, the system bridges the gap between complex AI capabilities and human healthcare needs.

### 3.2.1 MERITS

- **Instant Diagnosis:**

Users receive immediate diagnostic feedback from skin images, reducing delays in identifying potential conditions.

- **High Accuracy via Deep Learning:**

MobileNetV2 provides accurate classification based on trained features, ensuring reliable image analysis.

- **Educational Support through LLM:**

LLAMA 3 delivers informative, readable explanations about the diagnosed disease, improving user understanding.

- **User-Friendly Web Interface:**

Built with Gradio, the platform is intuitive and easy to navigate, even for non-technical users.

- **Improved Accessibility in Remote Areas:**

The system enables users from underserved regions to access dermatological insights without needing a physical clinic.

- **Reduces Clinical Workload:**

By offering a reliable self-assessment tool, the system eases pressure on healthcare providers and prioritizes critical cases.

## CHAPTER 4

### SYSTEM SPECIFICATION

#### 4.1 HARDWARE SPECIFICATION

- **Processor:** Minimum Intel Core i5 / AMD Ryzen 5 or higher
- **Storage:** 256 GB SSD (plus optional 1 TB HDD for dataset storage and backup)
- **Memory (RAM):** At least 8 GB (preferably 16 GB or more for smooth processing)

#### 4.2 SOFTWARE SPECIFICATIONS

- **Operating System:** Windows 10/11 or Ubuntu (Linux).
- **Programming Language & Libraries:** Python with TensorFlow, Keras, NumPy, pandas, and Matplotlib.
- **Deep Learning Model:** MobileNetV2
- **Generative AI Component:** LLAMA 3 via Ollama for disease description generation.
- **Development & Interface Tools:** Visual Studio Code, Jupyter Notebook, Gradio (UI), Git (version control)

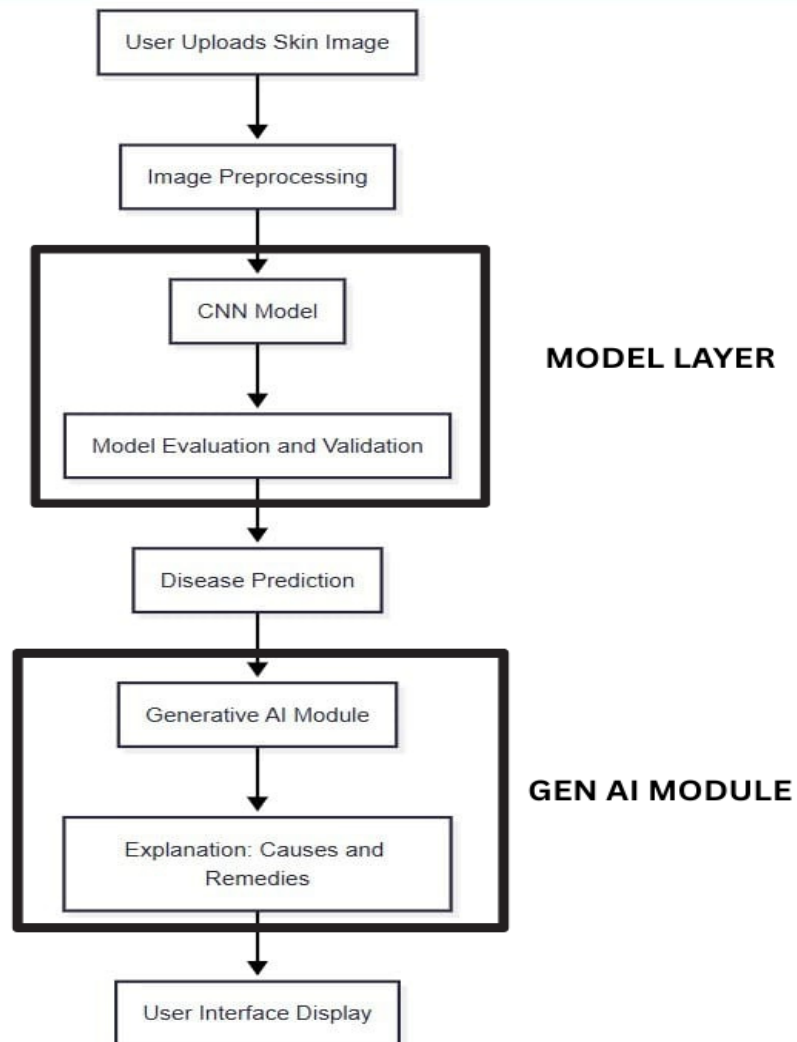
#### 4.3 FUNCTIONAL SPECIFICATIONS

- **Interactive User Interface:**  
  
Gradio-based UI enables real-time system interaction.
- **Image Classification Function:**  
  
MobileNetV2 detects skin diseases from images.
- **Text Generation Function:**  
  
LLAMA 3 generates simple disease explanations.
- **Educational Assistance:**  
  
Guides users with basic medical insights.

## CHAPTER 5

### SYSTEM DESIGN

#### 5.1 SYSTEM ARCHITECTURE



**Fig.5.1 System Architecture**

This architecture diagram represents an end-to-end AI-powered skin disease diagnosis system. The process begins with the user uploading a skin image, which is then passed through an Image Preprocessing stage. This stage handles essential tasks such as resizing, noise reduction, and normalization to prepare the image for analysis.

The preprocessed image enters the Model Layer, where a Convolutional Neural Network (CNN) is employed to extract deep features from the image. Following this, the system proceeds to Model Evaluation and Validation, ensuring the CNN's accuracy and robustness through performance metrics and test data.

Next, the system performs Disease Prediction based on the validated model outputs, classifying the skin condition accordingly. The result is then passed to the Generative AI Module, which forms part of the Gen AI Module. This module enhances the user experience by generating an Explanation of Causes and Remedies, providing understandable and actionable insights regarding the diagnosed condition.

Finally, the results and explanation are displayed on the User Interface, offering a complete diagnostic experience. This layered approach integrates traditional deep learning with generative AI, delivering not only classification but also interpretability and guidance, crucial for medical AI applications.

This architecture handles:

**User Uploads Skin Image:** Input image submitted for analysis.

**Image Preprocessing:** Resizes, denoises, and normalizes the image.

**Model Layer:**

- CNN Model: Extracts features from the image.
- Model Evaluation: Validates performance.

**Disease Prediction:** Classifies the skin condition.

**Gen AI Module:**

- Generative AI: Produces explanation.
- Explanation: Describes causes and remedies.

**User Interface Display:** Shows results and recommendations.

## **CHAPTER 6**

### **MODULE DESCRIPTIONS**

#### **6.1 DATA COLLECTION AND PREPROCESSING MODULE**

This module forms the backbone of the project by assembling and preparing the dataset for machine learning tasks. It ensures that the image data is clean, organized, and in the correct format to be fed into the CNN model. Quality preprocessing not only improves model accuracy but also prevents issues during training such as imbalance or noise in the data.

##### **6.1.1 Dataset Acquisition**

We utilized the HAM10000 dataset, a widely recognized dermatology dataset containing over 10,000 dermoscopic images. Each image is labeled with one of seven types of skin lesions such as melanoma, benign keratosis, and basal cell carcinoma. The dataset was obtained from Kaggle and includes metadata for diagnosis, localization, and patient demographics.

##### **6.1.2 Image Organization**

The original dataset was split into multiple parts (e.g., HAM10000\_images\_part\_1 and part\_2). These were merged into a single directory. Corresponding labels were extracted from a metadata CSV file and used to categorize the images into separate folders named by class. This class-wise folder organization is essential for directory-based loading during training.

##### **6.1.3 Data Preprocessing**

Each image was resized to  $224 \times 224$  pixels to match MobileNetV2's input layer dimensions. Pixel values were normalized to fall within the range of 0 to 1 by dividing by 255. The training data was augmented using rotation (up to 20 degrees), horizontal flips, zooming, and width/height shifts to simulate variability and reduce overfitting.

### 6.1.4 Dataset Splitting

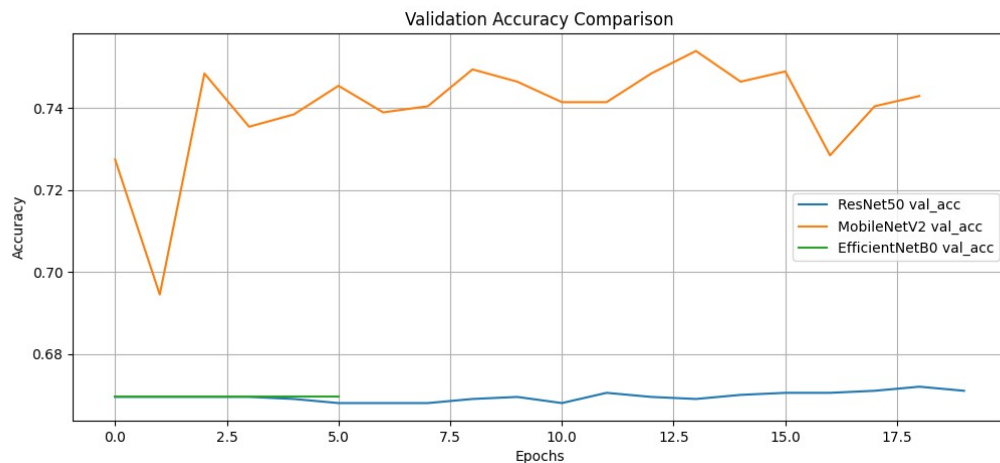
We performed an 80:20 stratified split to ensure balanced representation of each skin disease class in both the training and validation sets. Stratified splitting is critical in medical datasets where certain conditions may be underrepresented. This split was applied using Keras ImageDataGenerator to load and preprocess images dynamically during model training.

## 6.2 MODEL SELECTION MODULE

This module focuses on selecting an optimal deep learning architecture for multiclass skin disease classification. It involves analyzing existing models, evaluating their strengths and limitations, and adapting them to the dataset using transfer learning.

### 6.2.1 Model Architecture

In this project, we compared three CNN architectures—MobileNetV2, EfficientNetB0, and ResNet50—for skin disease classification. All models were fine-tuned using the same dataset and training parameters.



**Fig. 6.2.1 Model Selection**

MobileNetV2 outperformed the others, achieving the highest validation accuracy while maintaining low computational cost and faster training time. EfficientNetB0 and ResNet50 showed good performance but required more resources. Due to its efficiency and accuracy, MobileNetV2 was selected as the final model for implementation

### **6.2.2 Transfer Learning Strategy**

We used MobileNetV2 as a base model and added a custom classification head. The base layers were initially frozen during the first phase of training to retain generic feature extraction. Later, we unfroze selected deeper layers and fine-tuned the model on the skin dataset for domain-specific optimization.

### **6.2.3 Model Optimization**

The final layers included a GlobalAveragePooling2D layer, followed by dropout (for regularization), and a Dense layer with softmax activation to output probabilities across all skin disease classes. This head architecture ensures that only relevant features are passed to the output layer.

### **6.2.4 Tools and Frameworks**

TensorFlow and Keras were used as the primary frameworks for model selection and modification. Keras' modular API allowed for quick integration of pretrained models and tuning hyperparameters with minimal code complexity.

## **6.3 MODEL TRAINING MODULE**

This module describes the training of the CNN model, including configuration, monitoring, and validation. Proper training enables the model to learn meaningful features and accurately classify skin conditions.

### **6.3.1 Compilation Settings**

The model was compiled using the Adam optimizer due to its adaptive learning rate capabilities. Categorical cross-entropy was selected as the loss function since the problem is a multi-class classification task. Accuracy was chosen as the key metric for both training and validation monitoring.

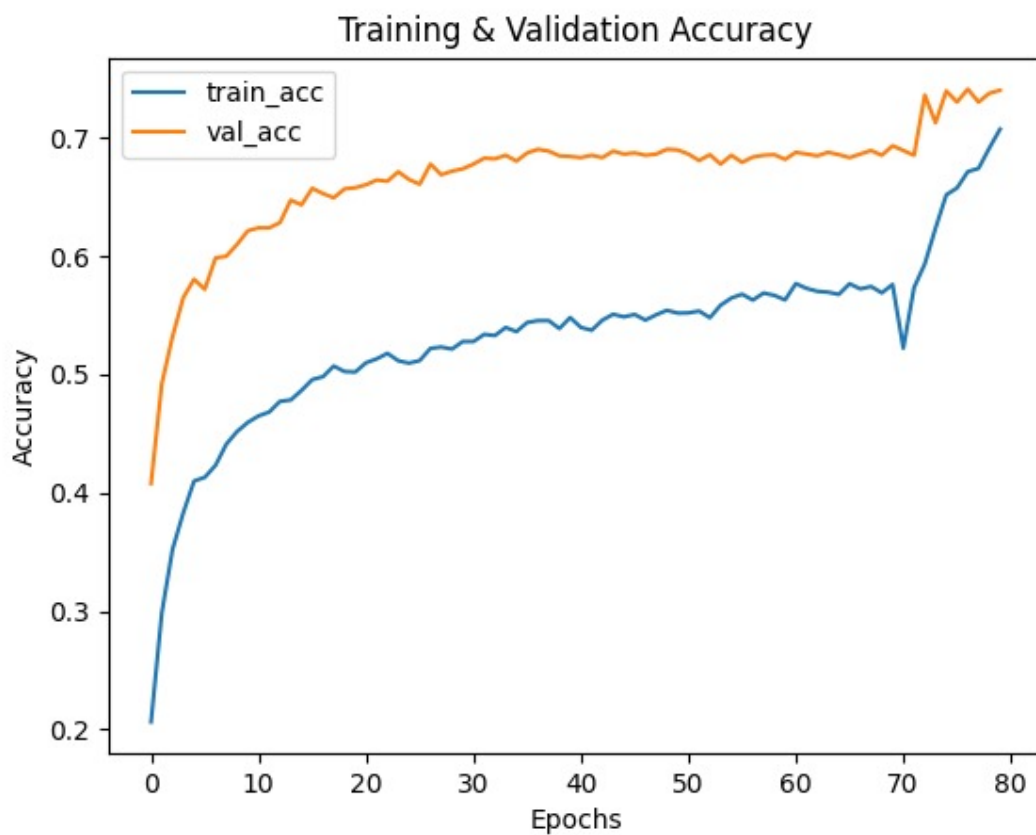
### **6.3.2 Training Configuration**

Training was performed for 70 epochs with a batch size of 32. Real-time augmentation was applied on-the-fly during training using ImageDataGenerator.

The model's weights were saved based on the best validation accuracy using ModelCheckpoint, and training was halted early using EarlyStopping if validation loss stopped improving.

### 6.3.3 Performance Evaluation

The model's accuracy and loss were tracked across each epoch using matplotlib visualizations. Additionally, a confusion matrix was plotted to analyze per-class performance. This helped us understand which diseases were frequently misclassified and informed future model refinements.



**Fig. 6.3.3 Model Training Result**

### 6.3.4 Model Persistence

The final trained model was saved in HDF5 (.h5) format. This preserved the architecture, learned weights, and training configuration, making it easy to reuse for prediction or deployment without retraining.

## **6.4 GENERATIVE AI MODULE**

This module leverages LLaMA 3 (via Ollama) to enrich the system's predictions with human-readable medical explanations, causes, and remedies. This enhances user understanding and provides added value beyond simple classification.

### **6.4.1 LLM Selection and Integration**

LLaMA 3, an open-source large language model, was chosen for its natural language generation capabilities and ease of local deployment using the Ollama runtime. The LLM was integrated with the CNN output to form a multi-modal AI pipeline.

### **6.4.2 Prompt Construction**

Once a prediction is made (e.g., "melanoma"), a pre-designed prompt such as "Explain the skin disease melanoma, its causes, and remedies in 3 paragraphs" is sent to the LLM. The output is generated in real-time and formatted as human-readable text.

### **6.4.3 Output Format**

The LLM returns three key segments: (1) a brief description of the disease, (2) common causes such as sun exposure or genetics, and (3) remedies or treatments, ranging from home care to dermatologist consultation. This improves the educational and assistive nature of the tool.

### **6.4.4 Ethical Considerations**

To avoid misuse, a disclaimer is added: "This is an AI-generated response and not a substitute for medical advice." This ensures users treat the tool as informative rather than diagnostic.

## **6.5 USER INTERFACE MODULE**

The final module focuses on delivering a seamless user experience. A user-friendly interface enables people with no technical background to interact with the system by uploading images and receiving understandable results.

### **6.5.1 Interface Framework**

The UI was built using Gradio, a Python library that simplifies the process of creating interactive web applications for machine learning models. Gradio supports fast prototyping and provides a drag-and-drop image upload feature.

### **6.5.2 Workflow**

The user uploads a skin image via the web interface. The backend MobileNetV2 model predicts the disease class, which is then passed to the LLaMA 3 model. The prediction and the generated explanation are shown in real-time on the same screen.

### **6.5.3 Design Features**

The interface features a simple layout with intuitive controls. Visual indicators like progress bars, result boxes, and markdown-based formatting enhance clarity. Preloaded sample images can be used for demonstration.

### **6.5.4 Accessibility and Extensibility**

The application can be hosted locally or deployed on the web. Future improvements may include multilingual support, audio feedback, or mobile optimization to make the tool even more accessible to the general public.

## **CHAPTER 7**

### **CONCLUSION AND FUTURE ENHANCEMENTS**

#### **7.1 CONCLUSION**

The project titled “Automated Skin Disease Diagnosis and Treatment Recommendations” demonstrates the powerful synergy between computer vision and generative ai in the domain of healthcare. By leveraging the mobilenetv2 convolutional neural network architecture, the system effectively classifies skin disease images into respective categories based on visual features. This lightweight yet robust cnn model was chosen for its efficiency and proven performance in mobile and embedded applications. Complementing the diagnostic model, the llama 3 generative model (via ollama) is used to generate informative, human-readable descriptions, including disease causes and general treatment recommendations.

This multimodal approach not only increases diagnostic accessibility for non-medical users but also enhances user engagement by providing context-rich explanations that a typical classifier would not offer. The integration of generative ai also allows the system to act as a first-level assistant that educates users about potential conditions while encouraging professional medical follow-up. Overall, the project successfully showcases how state-of-the-art pretrained models can be applied in a meaningful and socially beneficial application. It achieves a practical blend of accuracy, user understanding, and technical feasibility, making it a promising foundation for real-world implementations or further research in ai-driven dermatological tools.

## 7.2 FUTURE ENHANCEMENTS

The current system offers a strong foundation for image-based skin disease classification and AI-generated medical explanations. However, there are multiple areas where the system can be enhanced for improved performance, accuracy, and real-world usability. One of the most important improvements is to expand the training dataset. The current dataset, although effective, lacks diversity in skin tones, age groups, and lighting conditions. Increasing dataset size with high-resolution, diverse images from varied ethnic backgrounds will make the model more robust and reduce prediction bias. In addition to image data, incorporating patient metadata—such as age, gender, medical history, and geographical location—can greatly improve the system’s ability to provide context-aware, personalized predictions and remedies. This multimodal approach can help detect patterns specific to certain populations or regions.

On the generative AI side, while LLaMA 3 provides general explanations, fine-tuning it with domain-specific medical literature or replacing it with a specialized medical LLM would result in more accurate, fact-based, and medically reliable outputs. This would ensure greater trust in real-world applications. The system can also be enhanced by deploying it as a cross-platform mobile app or web application. Such deployment would allow wider access, particularly in rural or underserved regions. Adding features like voice commands, real-time image capture, and multi-language support would further increase accessibility. Finally, integrating a feedback loop where medical professionals can validate or refine predictions will lead to continuous learning and improved model accuracy over time. These future enhancements would transform the current academic prototype into a powerful, real-world tool for early detection and awareness in dermatology.

## **APPENDIX A**

### **SOURCE CODE**

```
#DATA PREPARATION

import pandas as pd

import os

import shutil

from sklearn.model_selection import train_test_split

# Paths

metadata_path = "./HAM10000_metadata.csv"

image_dir = "./images"

output_base_dir = "./split_data"

# Create output folders

train_dir = os.path.join(output_base_dir, "train")

val_dir = os.path.join(output_base_dir, "validation")

os.makedirs(train_dir, exist_ok=True)

os.makedirs(val_dir, exist_ok=True)

# Load metadata

df = pd.read_csv(metadata_path)

# Only keep images that exist in the folder

available_images = set(os.listdir(image_dir))

df = df[df['image_id'].apply(lambda x: x + ".jpg" in available_images)]

# Add .jpg extension
```

```

df['filename'] = df['image_id'] + ".jpg"

# Label is 'dx' column (diagnosis)

df['label'] = df['dx']

# Split the data

train_df, val_df = train_test_split(df, test_size=0.2, stratify=df['label'],
random_state=42)

#MODEL TRAINING

import os

import random

import numpy as np

import matplotlib.pyplot as plt

import tensorflow as tf

from tensorflow.keras import layers, models

from tensorflow.keras.preprocessing.image import ImageDataGenerator

from tensorflow.keras.applications import MobileNetV2

from tensorflow.keras.optimizers.schedules import CosineDecayRestarts

from tensorflow.keras.losses import CategoricalCrossentropy

from sklearn.utils.class_weight import compute_class_weight

from sklearn.metrics import classification_report, confusion_matrix

import seaborn as sns

import shutil

from tensorflow.keras.callbacks import ModelCheckpoint

```

```
# Set seed for reproducibility

SEED = 42

random.seed(SEED)

np.random.seed(SEED)

tf.random.set_seed(SEED)

# Define paths

TRAIN_DIR = '../dataset/split_data/train'

VAL_DIR = '../dataset/split_data/validation'

IMG_SIZE = (160, 160)

BATCH_SIZE = 8

NUM_CLASSES = 7

EPOCHS = 70

def build_partial_oversampled_generator():

    datagen = ImageDataGenerator(

        rescale=1./255,

        rotation_range=40,

        width_shift_range=0.2,

        height_shift_range=0.2,

        shear_range=0.2,

        zoom_range=0.2,

        horizontal_flip=True,

        fill_mode='nearest'
```

```

)

def generate_balanced_data(directory):

    all_images = []

    all_labels = []

    class_indices = {name: idx for idx, name in
enumerate(sorted(os.listdir(directory)))}

    for class_name in os.listdir(directory):

        class_path = os.path.join(directory, class_name)

        images = os.listdir(class_path)

        random.shuffle(images)

        limit = sample_limits[class_name]

        sampled_images = images[:limit] if len(images) > limit else images +
random.choices(images, k=limit - len(images))

        for img_name in sampled_images:

            all_images.append(os.path.join(class_path, img_name))

            all_labels.append(class_name)

    data = list(zip(all_images, all_labels))

    random.shuffle(data)

    all_images, all_labels = zip(*data)

    def generator():

        for img_path, label in zip(all_images, all_labels):

```

```

        img = tf.keras.preprocessing.image.load_img(img_path,
target_size=IMG_SIZE)

        img = tf.keras.preprocessing.image.img_to_array(img)

        img = datagen.random_transform(img)

        img = datagen.standardize(img)

        yield img, tf.keras.utils.to_categorical(class_indices[label],
num_classes=NUM_CLASSES)

dataset = tf.data.Dataset.from_generator(generator,output_types=(tf.float32,
tf.float32),output_shapes=([*IMG_SIZE, 3], [NUM_CLASSES]))

print(f"[INFO] Oversampled training images: {len(all_images)} from 7 classes.")

        return dataset.batch(BATCH_SIZE).prefetch(tf.data.AUTOTUNE)

    return generate_balanced_data(TRAIN_DIR)

# Build model with BatchNormalization and Dropout

base_model = MobileNetV2(weights='imagenet', include_top=False,
input_shape=(*IMG_SIZE, 3))

base_model.trainable = False

x = base_model.output

x = layers.GlobalAveragePooling2D()(x)

x = layers.BatchNormalization()(x)

x = layers.Dropout(0.4)(x)

x = layers.Dense(128, activation='relu')(x)

x = layers.BatchNormalization()(x)

x = layers.Dropout(0.3)(x)

```

```

outputs = layers.Dense(NUM_CLASSES, activation='softmax')(x)

model = models.Model(inputs=base_model.input, outputs=outputs)

# Focal loss implementation

def focal_loss(gamma=2., alpha=0.25):

    def focal_loss_fixed(y_true, y_pred):

        epsilon = 1e-7

        y_pred = tf.clip_by_value(y_pred, epsilon, 1. - epsilon)

        cross_entropy = -y_true * tf.math.log(y_pred)

        weight = alpha * tf.math.pow(1 - y_pred, gamma)

        loss = weight * cross_entropy

        return tf.reduce_mean(tf.reduce_sum(loss, axis=1))

    return focal_loss_fixed

# Compile model

model.compile(

    optimizer=tf.keras.optimizers.Adam(learning_rate=lr_schedule),

    loss=focal_loss(),

    metrics=['accuracy']

)

# Callbacks

checkpoint_cb = ModelCheckpoint(

    "best_skin_model.h5",

    monitor='val_accuracy',

```

```

        save_best_only=True,

        verbose=1

    )

# Train model

train_generator = build_partial_oversampled_generator()

history = model.fit(

    train_generator,

    epochs=EPOCHS,

    validation_data=val_generator,

    callbacks=[checkpoint_cb]

)

# Unfreeze and fine-tune

base_model.trainable = True

model.compile(

    optimizer=tf.keras.optimizers.Adam(learning_rate=lr_schedule),

    loss=focal_loss(),

    metrics=['accuracy']

)

fine_tune_epochs = 10

history_fine = model.fit(

    train_generator,

    epochs=fine_tune_epochs,

```

```

validation_data=val_generator,

callbacks=[checkpoint_cb]

)

model.save('mobilenetv2_model_fin2.h5')

print("Model saved.")

# Save plots

plt.figure()

plt.plot(history.history['accuracy'] + history_fine.history['accuracy'], label='train_acc')

plt.plot(history.history['val_accuracy'] + history_fine.history['val_accuracy'],
label='val_acc')

plt.title('Training & Validation Accuracy')

plt.xlabel('Epochs')

plt.ylabel('Accuracy')

plt.legend()

plt.savefig('training_progress_fin2.png')

#GEN AI EXPLANATIONS

import os

import numpy as np

import matplotlib.pyplot as plt

import tensorflow as tf

from tensorflow.keras.models import load_model

from PIL import Image

```

```

import subprocess

import json

from gen_ai.grad_cam import get_img_array, make_gradcam_heatmap,
save_and_display_gradcam

# === Constants ===

IMG_SIZE = (160, 160)

CLASS_NAMES = ['akiec', 'bcc', 'bkl', 'df', 'mel', 'nv', 'vasc']

MODEL_PATH = os.path.join('model', 'mobilenetv2_model_fin2.h5')

OUTPUT_DIR = 'gen_ai/gen_ai_outputs'

os.makedirs(OUTPUT_DIR, exist_ok=True)

def process_image_and_generate_response(image_path):

    # === Load Model ===

    print("[INFO] Loading model...")

    model = load_model(MODEL_PATH, compile=False)

    print("[INFO] Model loaded successfully.")

    # === Preprocess and Predict ===

    print(f"[INFO] Processing image: {image_path}")

    img_array = get_img_array(image_path, size=IMG_SIZE)

    # Predict class

    preds = model.predict(img_array)

    pred_class_idx = np.argmax(preds[0])

    pred_class = CLASS_NAMES[pred_class_idx]

```

```

confidence = preds[0][pred_class_idx] * 100

print(f"[INFO] Predicted class: {pred_class} ({confidence:.2f}%)")

# === Prepare Prompt for LLM (Ollama) ===

gradcam_summary = "Red region is the dominant focus area in the lesion core."

prompt_data = {

    "image_path": os.path.basename(output_path),

    "prediction": pred_class,

    "summary": gradcam_summary

}

prompt = f"""

You are a medical assistant AI helping explain skin disease diagnoses.

Based on the following information:

- Predicted Disease: {prompt_data['prediction']}

- Grad-CAM Summary: {prompt_data['summary']}

Provide a clear and concise explanation in the following format:

About the Disease:

...

Causes:

...

Remedies:

...

Keep the content under 200 words total.

```

```

"""

# === Call Ollama LLM ===

print("[INFO] Generating explanation using Ollama...")

try:

    result = subprocess.run([

        "ollama", "run", "llama3", prompt

    ], capture_output=True, text=True, check=True, encoding="utf-
8", errors="replace")

    llm_response = result.stdout.strip()

except subprocess.CalledProcessError as e:

    print("[ERROR] Ollama failed to generate response.")

    llm_response = "[LLM ERROR] Could not generate explanation."

# === Save LLM Output ===

llm_output_path = os.path.join(OUTPUT_DIR,
f"llm_output_{os.path.basename(image_path).split('.')[0]}.txt")

with open(llm_output_path, 'w') as f:

    f.write(llm_response)

print("[INFO] Explanation from LLM:")

print(llm_response)

print(f"[INFO] Saved to: {llm_output_path}")

return {

    "pred_class": pred_class,

```

```

        "gradcam_path": output_path,

        "llm_output": llm_response

    }


#MAIN

import gradio as gr

from gen_ai.gen_ai_model import process_image_and_generate_response

# Diagnose Function

def diagnose(image):

    result = process_image_and_generate_response(image)

    return (

        result["pred_class"],

        result["conf_score"],

        result["gradcam_path"],

        result["llm_output"],

        "✔ Diagnosis complete."

    )

def loading_status():

    return "❏ Diagnosing... please wait..."

# App UI

with gr.Blocks(css=custom_css, theme=gr.themes.Soft()) as app:

```

```

# Landing Page

with gr.Column(visible=True) as landing_section:

    with gr.Row(elem_classes="hero"):

        with gr.Row(elem_classes="features"):

            gr.HTML("""

                <div class="feature-box">

                    <h3>Easy Image Upload</h3>

                    <p>Upload a dermoscopic image from your device and let the model
analyze it in seconds.</p>

                </div>

            """)

            gr.HTML("""

                <div class="feature-box">

                    <h3>LLM Explanations</h3>

                    <p>Local LLM provides a short, clear explanation of the disease, causes,
and remedies.</p>

                </div>

            """)

get_started_btn = gr.Button("🚀 Get Started", elem_id="get-started")

```

```

# Diagnosis Section

with gr.Column(visible=False) as diagnosis_section:

    gr.Markdown("<div id='title'>🩺 Skin Disease Diagnosis AI</div>")

    gr.Markdown("<div id='description'>Upload a dermoscopic image to detect and explain possible skin diseases using AI.</div>")

    with gr.Row():

        with gr.Column():

            image_input = gr.Image(label="Upload Skin Image", type="filepath")

            submit_btn = gr.Button("🔍 Diagnose")

        with gr.Column():

            heatmap_output = gr.Image(label="Grad-CAM Heatmap")

    with gr.Row():

        with gr.Column():

            pred_output = gr.Textbox(label="Predicted Class", interactive=False)

        with gr.Column():

            conf_output = gr.Textbox(label="Confidence Score", interactive=False)

            llm_output = gr.Textbox(label="Explanation (Generated by LLM)", lines=6,
            interactive=False)

            status_output = gr.Markdown("")

            gr.Markdown("<div id='disclaimer'>⚠️ This tool is for educational and research purposes only. It is not intended for medical diagnosis or treatment.</div>")

# Button Behavior

```

```

submit_btn.click(fn=loading_status, outputs=[status_output], queue=False)

submit_btn.click(

    fn=diagnose,

    inputs=[image_input],

    outputs=[pred_output, conf_output, heatmap_output, llm_output,
status_output],

    show_progress=True

)

def toggle_sections():

    return gr.update(visible=False), gr.update(visible=True)

get_started_btn.click(

    fn=toggle_sections,

    outputs=[landing_section, diagnosis_section]

)

# Launch the app

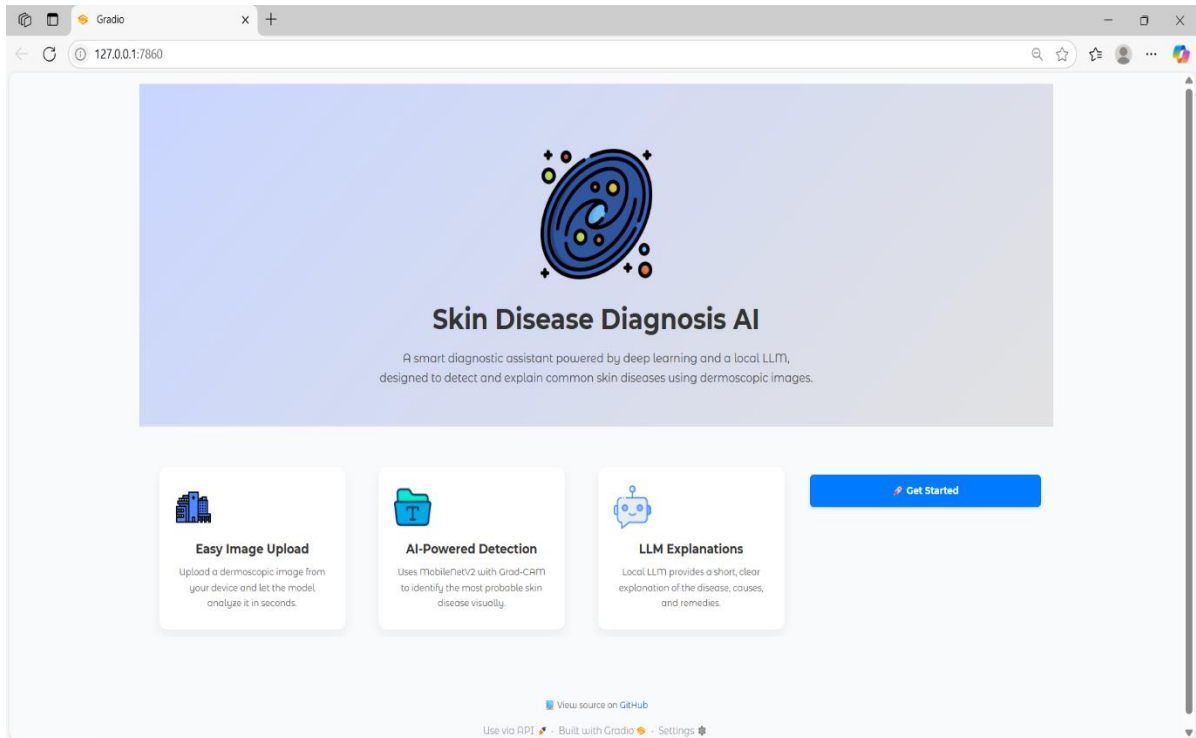
if __name__ == "__main__":

    app.launch()

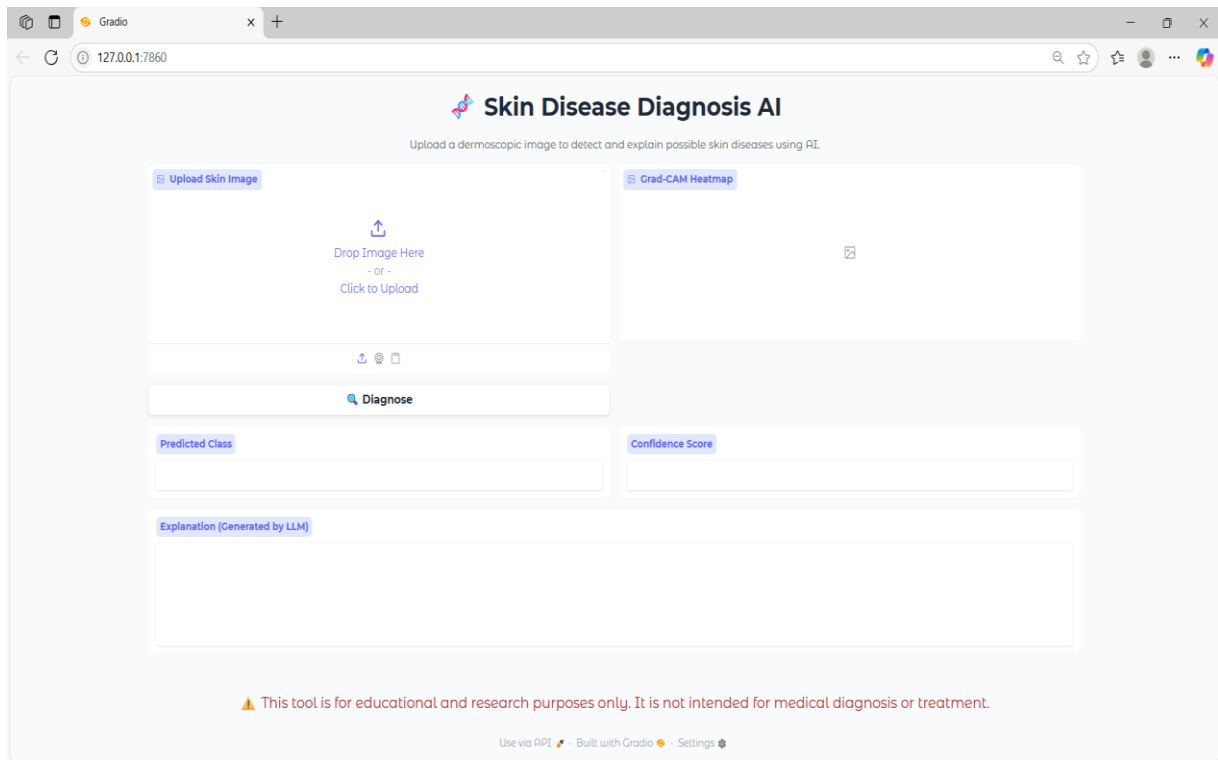
```

## APPENDIX B

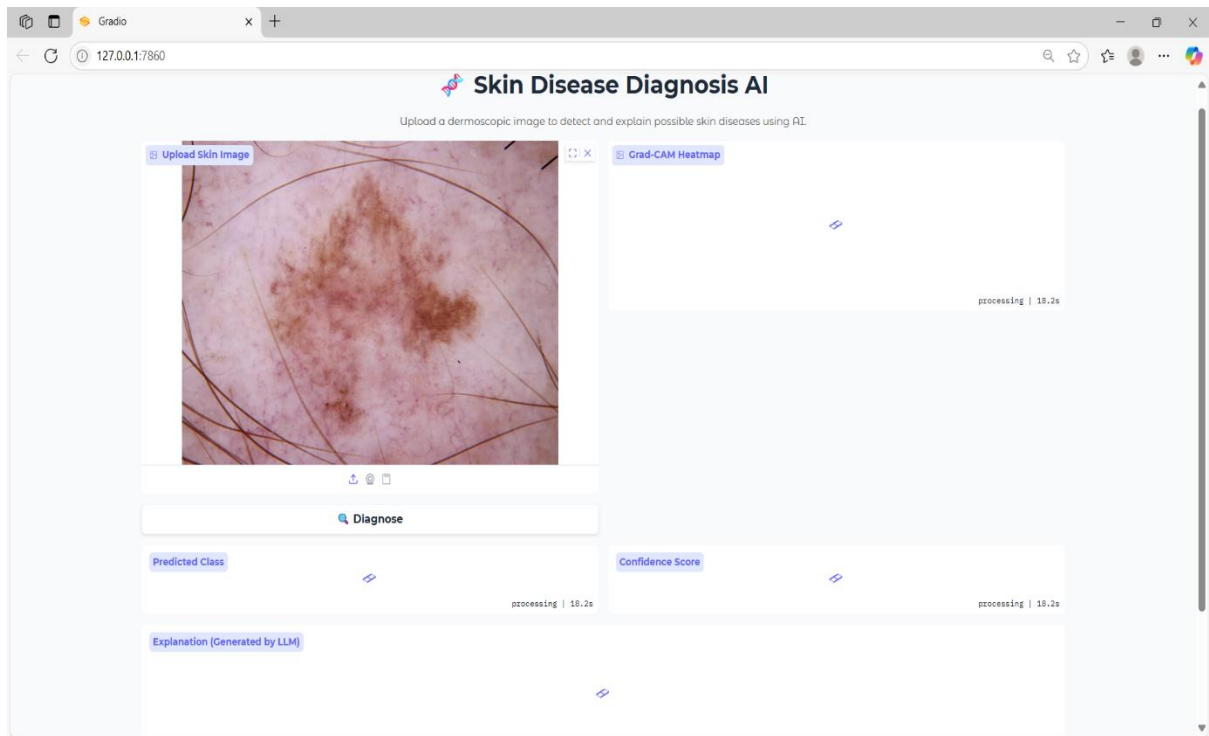
### SCREENSHOTS



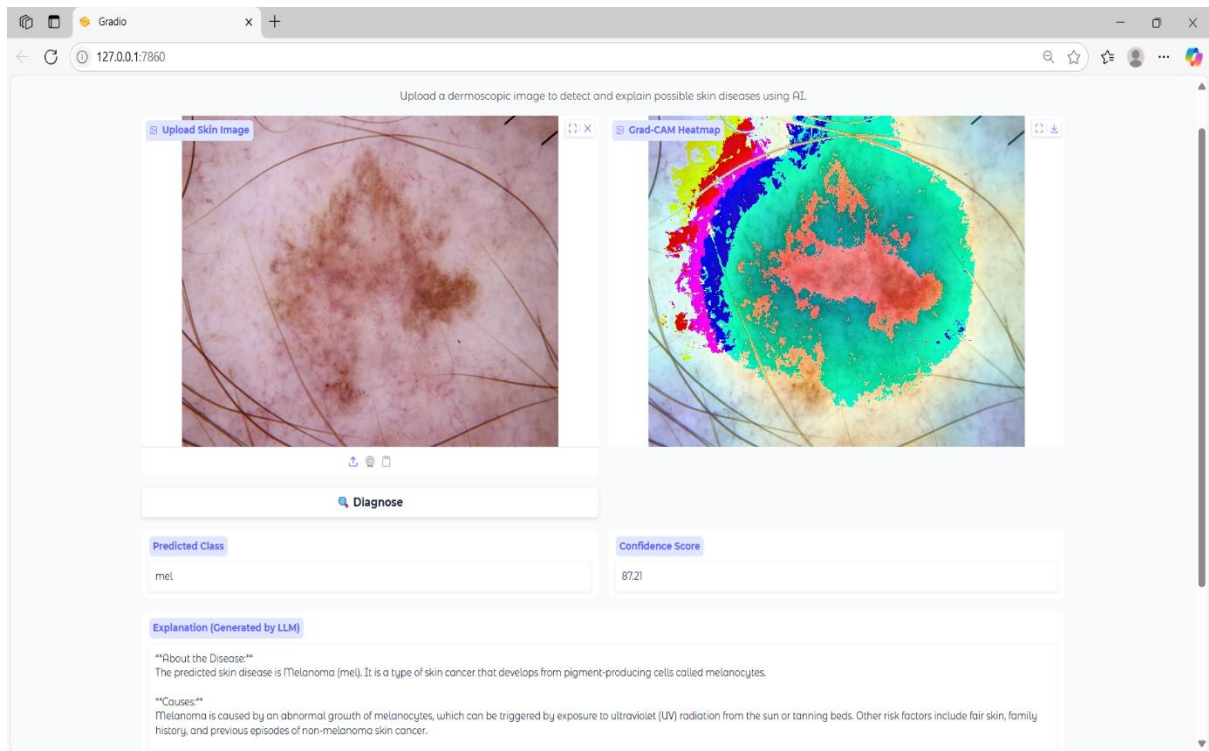
**Fig. B.1 Home Page**



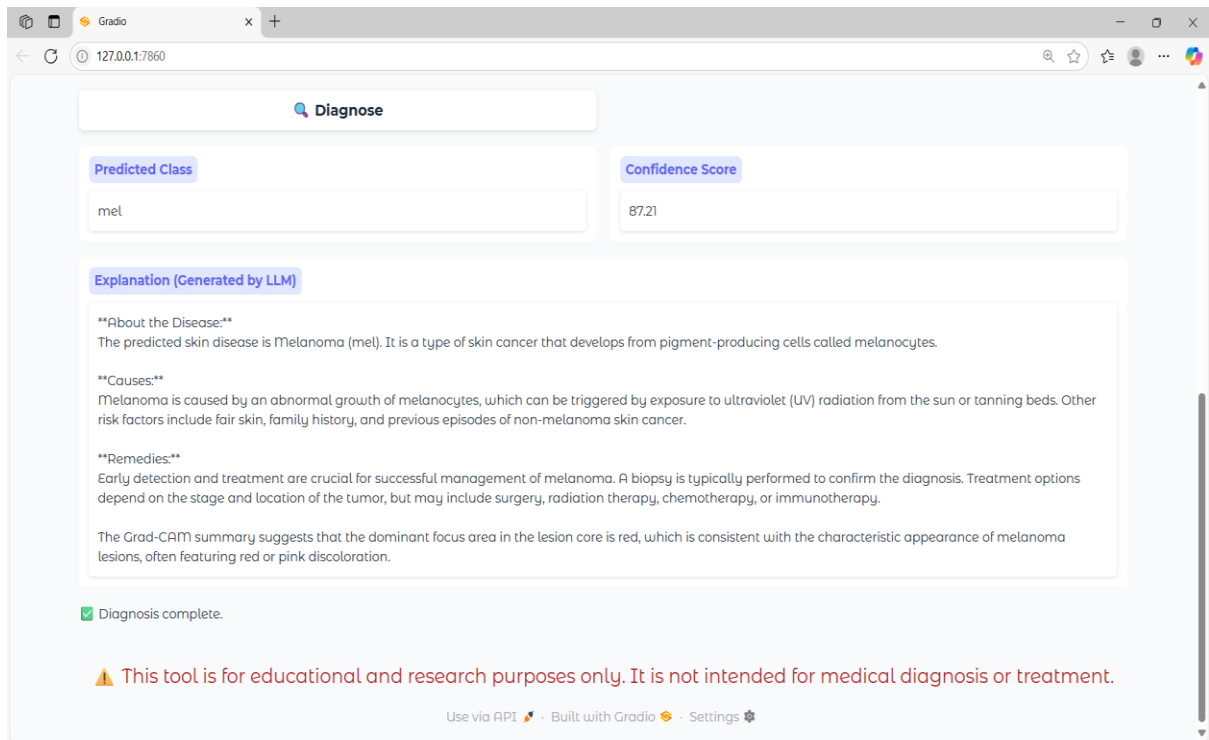
**Fig. B.2 User Interface**



**Fig. B.3 Image Processing**



**Fig. B.4 Predictions and Results**



**Fig. B.5 LLM Explanations**

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